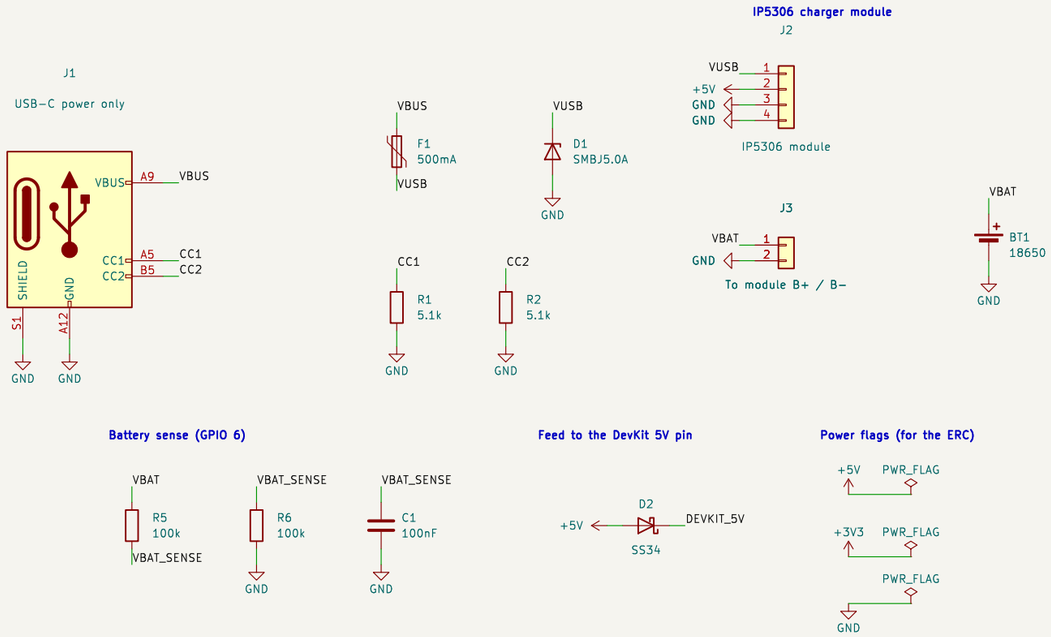
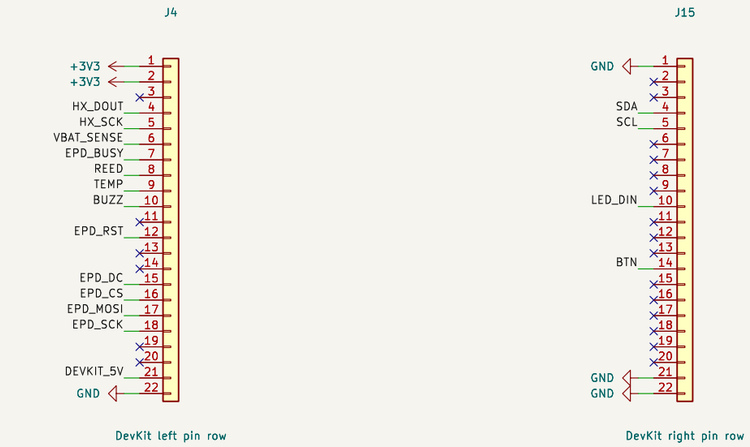
EcoTwin Smart Bin — Brain Pod carrier board (95 x 60 mm, 2 layers)

Team VK26-IITI-AHM-0136 | IIT (BHU) Varanasi | Vishwakarma Awards 2026-27

1 Power: USB-C in, charger module, battery, battery sense

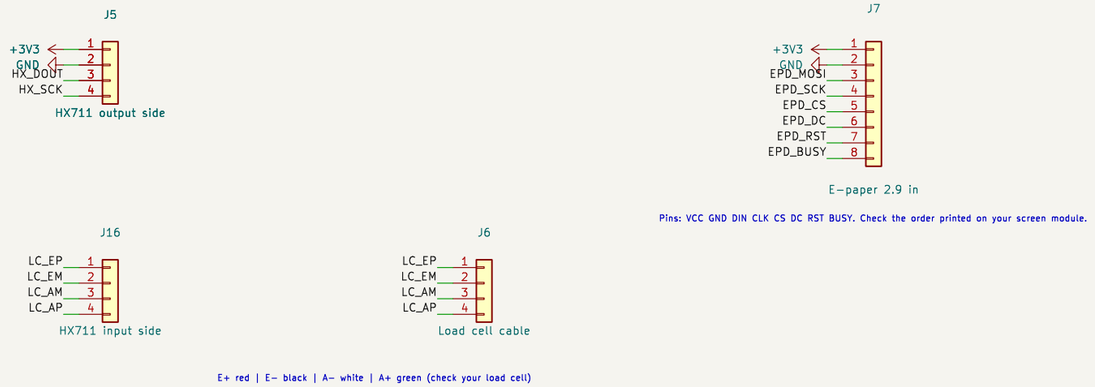


2 ESP32-S3-DevKitC-1 sockets

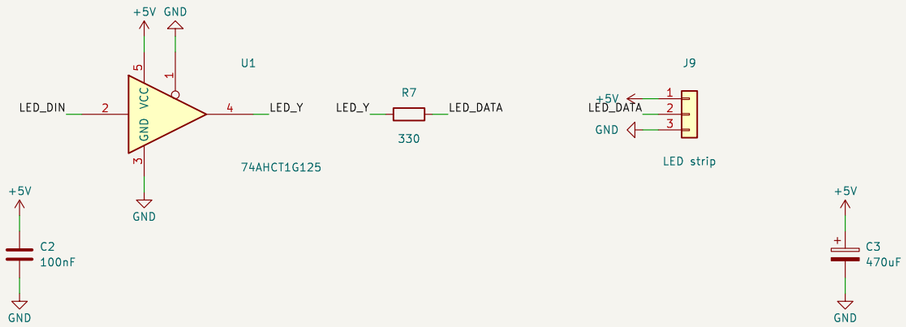


Pin order follows the ESP32-S3-DevKitC-1 board. Check it against the real board before ordering.

3 Weighing (HX711 + load cell) and e-paper display



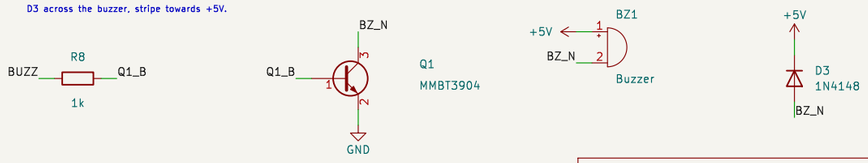
4 Status light (WS2812 strip, GPIO 38)



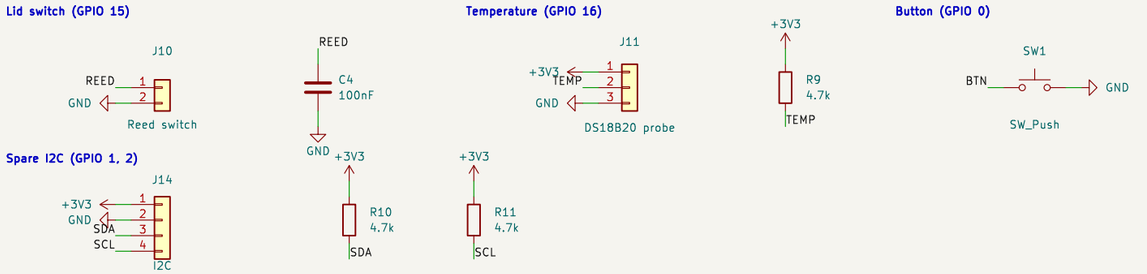
C2 sits next to U1. C3 sits across the strip supply.

5 Buzzer (GPIO 17)

D3 across the buzzer, stripe towards +5V.



6 Lid switch, temperature probe, button, spare I2C



IIT (BHU) Varanasi — team VK26-IITI-AHM-0136		
Sheet: /		
File: EcoTwin_Board.kicad_sch		
Title: EcoTwin Smart Bin carrier board		
Size: A3	Date: 2026-10-22	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1